

Urban Air Quality & Public Health Impact Analyzer

Analyzing 2,732 daily readings across 30 major Indian cities (Jan 2022– Dec 2023) to uncover pollution patterns, seasonal trends, and health outcomes for data-driven urban planning.



Dataset at a Glance



Key features: AQI, PM2.5, PM10, NO₂, SO₂, CO, O₃, NH₃, weather conditions, hospital admissions, respiratory & cardiovascular cases.

2,732 Rows

After cleaning from
3,000 raw records

33 Columns

Location, pollutants,
weather & health data

30 Cities

Across 18 Indian states

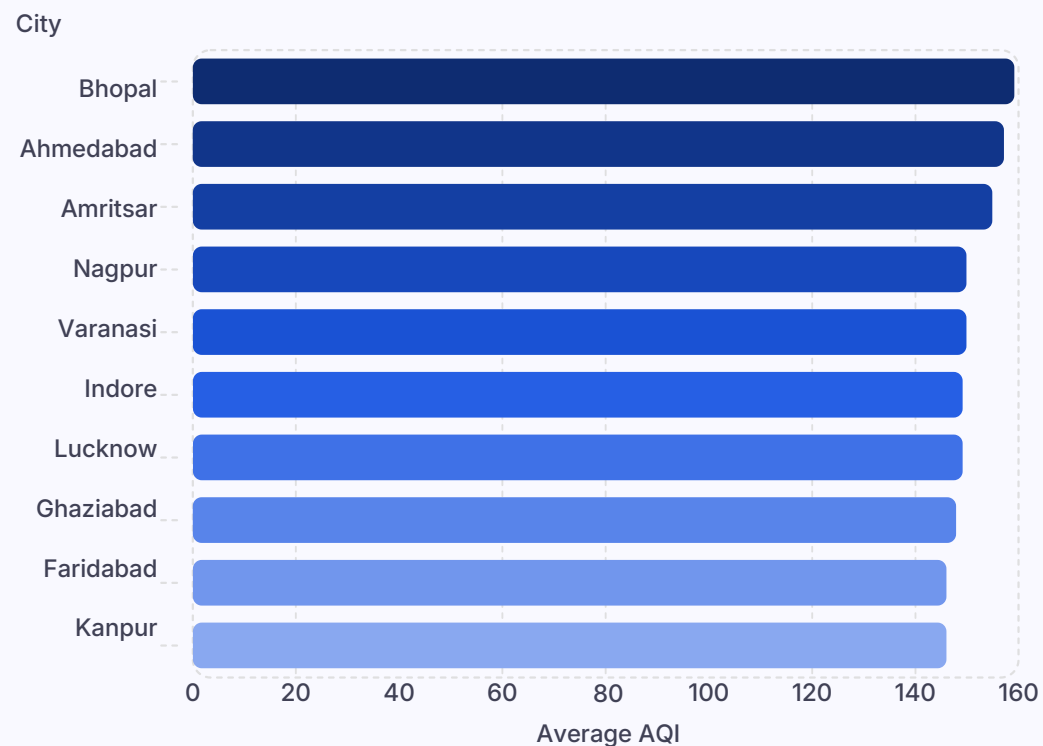
~4% Missing

In AQI, PM2.5,
temperature &
admissions

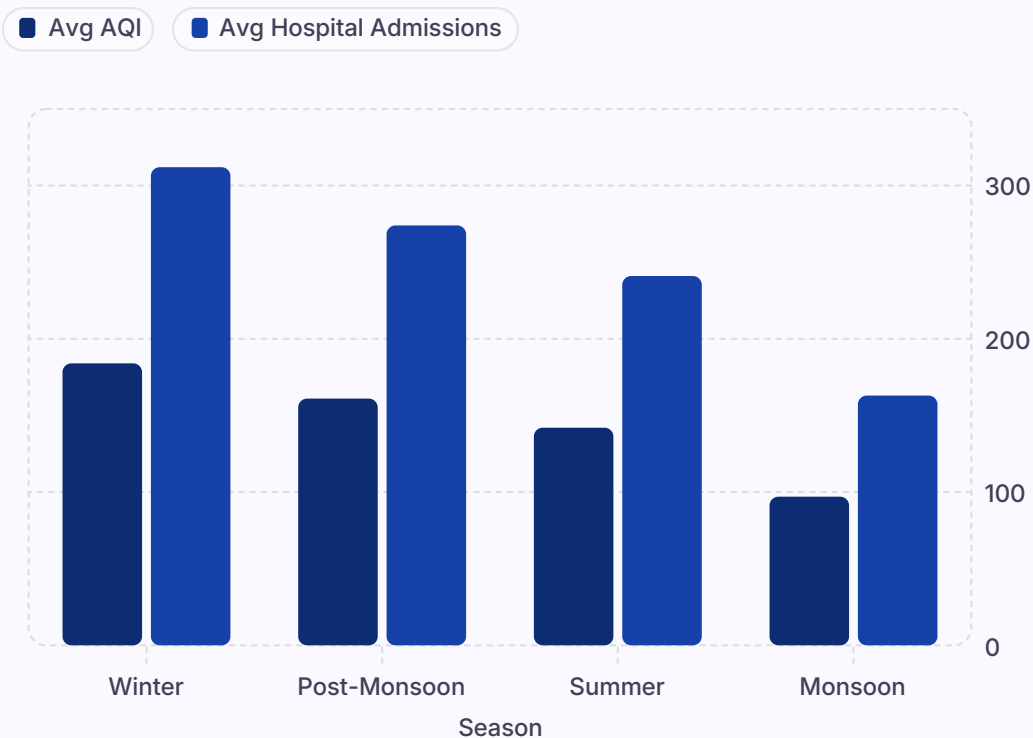
AQI Distribution & Top Polluted Cities

AQI Distribution

Right-skewed distribution — most readings fall between 100–160 AQI. Mean: 142.6, Median: 134.8.



Seasonal Air Quality Analysis



Winter is the Worst Season

Winter AQI peaks at **184.3** — nearly **2× Monsoon levels**. Hospital admissions follow the same trend, spiking to **312** in Winter vs. **163** in Monsoon.

Key Insight: PM_{2.5}, NO₂, and SO₂ all peak in Winter, directly correlating with respiratory and cardiovascular case surges.

AQI vs. Public Health Impact

142.56

Across all 30 cities

564K

Hospital Admissions

Total across study period

37.55

Score (0–100scale)

63

Hazardous Days

Recorded across all cities

Hospital admissions rise sharply with AQI category: **82** (Good) → **521** (Hazardous) — a **6× increase**. Uttar Pradesh recorded the most hazardous days (**18**), followed by Madhya Pradesh (**14**) and Punjab (**11**).

Power BI Interactive Dashboard



Dashboard Components

- **KPICards:** AvgAQI, Total Admissions, Health Risk, Cities, Hazardous Days
- Monthly AQI Trend: 2022 vs 2023 comparison line chart
- PM2.5 vs Hospital **Admissions:** Scatter plot with AQI color-coding
- Pollutant Heatmap: Seasonal breakdown of all 7 pollutants
- Interactive Slicers: Filter by City, Season, Year, AQI Category

Key Recommendations



Target Winter Interventions

Restrict vehicles & industry Nov–Feb when AQI exceeds 180. Activate alert systems.



Focus on Industrial Zones

Bhopal, Ahmedabad & Amritsar show highest pollution. Enforce stricter emission controls and green buffer zones.



Strengthen Healthcare

Scale hospital capacity in Winter & Post-Monsoon. Boost respiratory and cardiovascular resources.



Increase Green Cover

Target minimum **25% greencover** in Industrial and Urban zones to lower AQI.



Promote Public Transport

Weekday pollution exceeds weekends. Subsidized transit can reduce NO₂ and CO emissions.



Expand AQI Monitoring

Increase monitoring stations in Tier-2 cities to close data gaps and improve early warnings.